

● ● Marketing analytics case study · Google Data Analytics capstone

Where should the Google Merchandise Store spend its next marketing dollar?

Channel attribution, a retargeting model and two tests that would settle what attribution can't

Google Analytics 360 export

Aug 2016 – Aug 2017

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Analysis designed and directed by me; code written with AI assistance.

Stock photo



Five findings for next quarter's budget

Up to 15 pts

of purchase credit GA's report likely gives Organic Search that return visits earned

34%

of purchases come from visitors returning on their own (44% of capped revenue); when they first came from a campaign, GA's report credits that campaign

\$0.78–1.21

is the most a Paid Search click is worth at a 50% margin, if every sale needed the ad

71%

of later buyers are in the model's top 10%, yet retargeting them is worth only up to about \$690–745 a month before ad costs

89%

of the revenue GA credits to Display came from one corporate buyer: hold Display's budget and test it on site visits

● ● Before any analysis

41%

of the store's revenue came from Google employees, not marketing

In the public data these visits appear as "Referral" with the source hidden. Google's training copy shows the source, mall.googleplex.com, the employee store link.

5%

of visitors

47%

of purchase sessions

99.2%

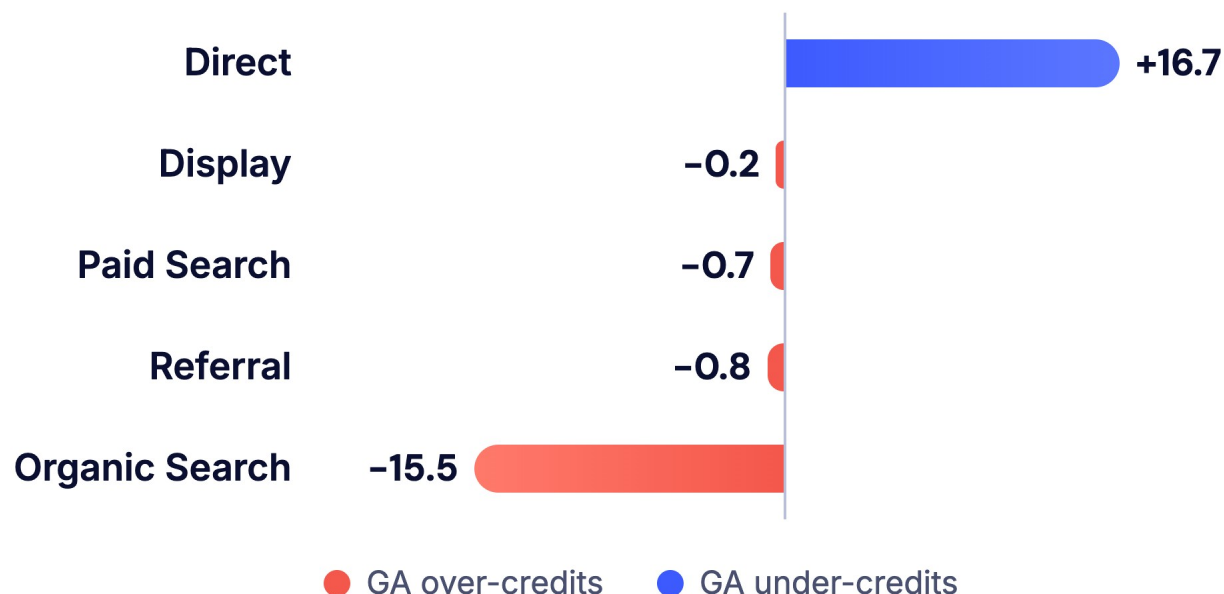
precision of my employee
filter against that copy

98.2%

of employee purchases the
filter catches

GA likely over-credits Organic Search by up to 15 points

Share of purchases a data-driven model credits, minus the share GA's report credits, in points, if every direct return was self-initiated



How big is the gap?

15.5

points if every direct return was self-initiated
(95% CI 14.4–16.6)

13–15

points at a benchmark from how returning
visitors behave

1.8

points on GA's own labels

●● Channel budget: what a paid click is worth

A Paid Search click's value is a ceiling, not a profit

Bid cap per Paid Search click at a 50% gross margin

If the ads caused...	Bid cap per click
every attributed sale	\$0.78–\$1.21
half of them	\$0.39–\$0.61
a quarter of them	\$0.19–\$0.30

A click is credited with \$1.56–\$2.43 of revenue under seven attribution rules; the cap is half of that times the share the ads caused.

65 of 65

Paid Search purchases with a readable keyword came on brand keywords

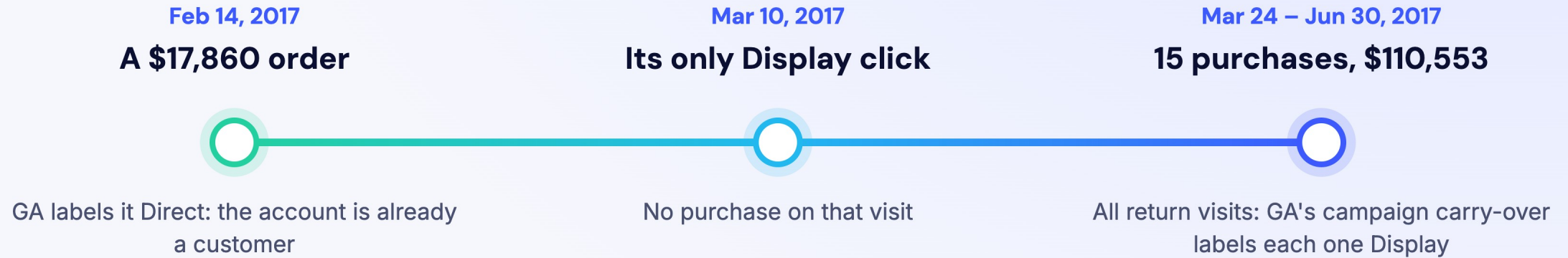
231 of 437

buying journeys with Paid Search touch no other channel, so five of the seven rules give it full credit



Split brand from non-brand campaigns, and test pausing brand ads, before raising any bid.

One corporate buyer made Display look like a winner



89%

of the revenue GA credits to Display is this one account

\$2.84–\$3.87

a Display click's attributed value without it, under every rule

Recommendation

Report the account on its own, like employees, and test Display on site visits.

The model's top 10% holds 71% of later buyers



Later buyers reached by targeting the top 10%

My model (gradient boosting)



Two-line rule: North America, then funnel step



Random targeting



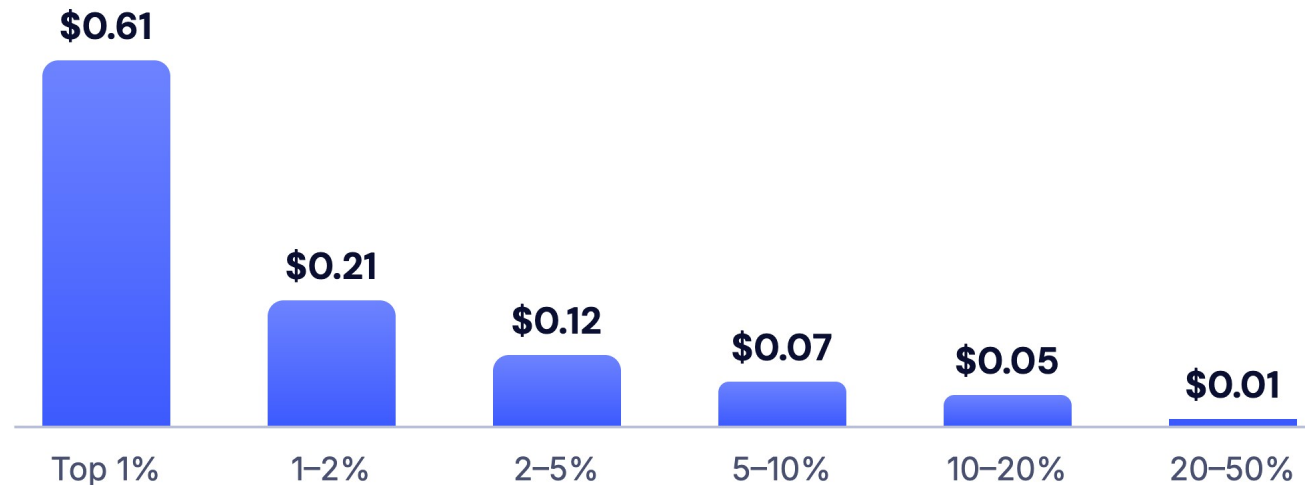
What predicts a return purchase (training months)

A North American first visit (**15.7×** the odds),
adding to cart (**3.7×**), and arriving by
Paid Search or Display (about **15–21×** Social's odds)

●● Retargeting: is it worth it?

Only top-scored visitors are worth paying to bring back

The most worth paying per visitor the ads actually reach, for 30 days, by score band



Staff flagged using the whole year; without hindsight, each band moves a few cents

\$690–745

a month at most in extra gross profit from the top 10%, if the ads reach everyone (before ad costs)

\$920–950

a month at most for the top 20%

\$207–\$1,936

a month at most for the top 10% at lifts of 5–20% and margins of 30–70%

Eight actions: four now, two tests, two to explore

Now

Report employees and the key account as their own segments

Web analytics
41% of revenue; \$128k

Show a multi-touch view beside GA's channel report

Marketing, Finance
up to 15 pts of credit

Split brand from non-brand search; bid under the ceiling

Paid media
\$1.56–2.43 per click

Review any spend on YouTube promotion and affiliates

Paid media
Oct–Nov: 98k visits, no sales

Test

Retarget only top-scored visitors; 50/50 holdout test for 12 months

Retargeting lead
Up to \$690–950 a month

Hold Display's budget; run a 12-week site-visit holdout

Paid media
\$17k revenue a year

Explore

Retention: email and reminders for past visitors

Head of Marketing
34% of purchases

A direct sales path for corporate buyers

Marketing, Sales
\$249k bulk purchases

Four decisions for today

1 Report employees and the key account on their own?

Then every channel number matches this analysis. Employees alone are 41% of revenue.

2 Run a 12-month, 50/50 retargeting test of the top 20%?

The only design that can measure the lift: it detects a lift of 14% or more; 10% would need about 23 months.

3 Hold Display's budget and test its own audience on site visits for 12 weeks?

About 2 purchases a week are too few to test. Visits work if that audience makes at most about 500–1,000 visits a week.

4 Split brand from non-brand search and test pausing brand ads?

Every readable keyword behind a Paid Search purchase was a brand keyword (the store's or Google's name).

How sure I am

● High confidence

Employee traffic, checked against Google's unredacted data

The key account's timeline, from its own visits

The retargeting ranking, tested on unseen months: 7–10 points ahead of the two-line rule

● Moderate confidence

Organic Search's over-credit: the direction is clear, the size rests on an assumption

The returning-visitor share (34%), for the same reason

Retargeting's value: it assumes a 10% lift, a 50% margin and full reach

● Low confidence

Whether Paid Search clicks are incremental: every purchase with a readable keyword came on a brand keyword

What Display adds

The retargeting lift, borrowed from published experiments

→ Attribution describes paths, not causes. The recommended tests are designed to settle the rest.

The teaching lab's model mostly finds employees

Share of first-time visitors who are Google employees, by the lab model's predicted chance of a later purchase (May–Jun 2017)



61%

of the future buyers the lab model learns from are Google employees

2 of 1,020

top-ranked visitors are outside customers who later bought

On outside visitors, the lab model scores 0.86, not 0.91

ROC-AUC of BigQuery ML models on the lab's evaluation months, May–Jun 2017

Model	All first visits	Outside visitors
The lab's model, as published	0.910	0.863
Without source, medium and channel	0.885	0.882
Trained on outside visitors only	—	0.877

ML.EVALUATE reports 0.909 for the recreated model (published 0.91); recomputed from row-level predictions it is 0.910.

–0.047

drop in its ROC-AUC on outside visitors
(95% CI -0.059 to -0.035)

59–66%

of the top 1% are still employees in
both audit versions, through geography
and engagement



Fine for teaching; take internal traffic out before using a model like this to target anyone.

How I did it

Data and SQL

903,653 sessions queried in BigQuery, with nested page hits unpacked for funnel steps

Google employees and one key account reported as their own segments

A keyword extract (0.25 GB) to split brand from non-brand search

Models and tests

Chi-square, z-tests and logistic regression with odds ratios

Logistic regression, random forest and gradient boosting, tuned on earlier months

Seven attribution rules, including a third-order Markov chain

Power analysis for both recommended tests

Validation

Tested on months the models never saw in training or tuning

Bootstrap intervals that resample visitors

Over 200 automated tests; a fresh clone runs end to end

Google's lab recreated in BigQuery ML from its own SQL